1. Variables Relationship

Exercise 1: Scatter Plot with Regression Line

- Dataset: `tips` dataset from Seaborn.
- Task: Visualize the relationship between `total_bill` and `tip`. Add a regression line using Seaborn's `regplot`.
- Challenge: Highlight data points for male and female customers using different colors.

Exercise 2: Pairwise Relationships

- Dataset: `iris` dataset from Seaborn.
- Task: Create a pairplot to explore the relationships between all numerical features (`sepal_length`, `sepal_width`, `petal_length`, and `petal_width`).
- Challenge: Use different colors for species and analyze patterns.

Exercise 3: Heatmap for Correlation Matrix

- Dataset: Load any dataset with numerical features (e.g., `penguins` from Seaborn).
- Task: Compute and visualize the correlation matrix using Seaborn's `heatmap`.
- Challenge: Annotate the heatmap to show the exact correlation coefficients.

2. Comparisons

Exercise 1: Bar Plot

- Dataset: `diamonds` dataset from Seaborn.
- Task: Compare the average price of diamonds based on their cut (e.g., `Ideal`, `Premium`).
- Challenge: Include error bars to show the standard deviation.

Exercise 2: Box Plot

- Dataset: `tips` dataset from Seaborn.
- Task: Create a box plot to compare `total_bill` for different days of the week.
- Challenge: Overlay a swarm plot to show individual data points.

Exercise 3: Violin Plot

- Dataset: `penguins` dataset from Seaborn.
- Task: Visualize the distribution of `flipper_length_mm` for different species using a violin plot.
- Challenge: Split the violins by `sex`.

3. Time-Series Data

Exercise 1: Line Plot

- Dataset: Any time-series dataset (e.g., `flights` dataset from Seaborn).
- Task: Plot the number of passengers over time.
- Challenge: Use different line styles and markers for each year.

Exercise 2: Rolling Mean and Trends

- Dataset: Custom dataset or generated using Python.
- Task: Simulate daily stock prices, then plot the rolling mean (7-day and 30-day) using Matplotlib.
- Challenge: Highlight peaks and troughs in the data using annotations.

Exercise 3: Seasonal Heatmap

- Dataset: `flights` dataset from Seaborn.
- Task: Create a heatmap showing the number of passengers for each month across years.
- Challenge: Highlight months with the highest passengers for each year.

4. Additional Interesting Exercises

Exercise 1: Nested Pie Chart

- Dataset: Custom dataset (e.g., proportions of expenses across categories and subcategories).
- Task: Visualize the data using a nested pie chart in Matplotlib.
- Challenge: Use color palettes to distinguish subcategories.

Exercise 2: Joint Distribution

- Dataset: `penguins` dataset from Seaborn.
- Task: Visualize the joint distribution of `bill_length_mm` and `bill_depth_mm` using Seaborn's `jointplot` with different kinds (`scatter`, `hex`, `kde`).
- Challenge: Add marginal histograms.

Exercise 3: Interactive Plot

- Dataset: Simulate or use a real dataset (e.g., COVID-19 case trends).
- Task: Create an interactive line plot showing cases over time for multiple regions using Matplotlib and the `mpl_toolkits` library.
- Challenge: Allow dynamic highlighting of specific regions.

Exercise 4: Animated Visualization

- Dataset: Simulate data (e.g., a moving particle or stock prices).
- Task: Create an animation of data points evolving over time using Matplotlib's `FuncAnimation`.
- Challenge: Add labels showing the current data value during the animation.